

Background on Countable Borel Relations

We will always assume that X is a standard Borel space, possibly equipped with a probability measure μ .

Definition:

- If $\Gamma \curvearrowright X$ by Borel automorphisms, then the orbit equivalence relation E is defined by taking $(x, y) \in E$ if and only if there is $\gamma \in \Gamma$ such that $\gamma.x = y$. We write E_Γ^X .
- We say an equivalence relation E on X is finite/countable if the equivalence classes are finite/countable.
- Given an equivalence relation E , we let $[E]$ denote the group of Borel automorphisms of X whose graphs are contained in E (known as the *full group*). The set $[[E]]$ consists of partial Borel isomorphisms $\phi: A \rightarrow B$ whose graphs are contained in E ; we write $A = \text{dom}(\phi)$ and $B = \text{range}(\phi)$.
- We say a probability measure μ is E -invariant if for all $f \in [E]$, we have $f_*\mu = \mu$. Equivalently, for all $\phi \in [[E]]$, we have that $\mu(\text{dom}(\phi)) = \mu(\text{range}(\phi))$.
- We say a measure μ is E -ergodic if every E -invariant Borel set is null or conull.
- We say (X, E) and (Y, F) are Borel isomorphic if there is a Borel bijection $\pi: X \rightarrow Y$ such that for all $x, y \in X$, we have xEy if and only if $\pi(x)F\pi(y)$.
- We say Borel actions $\Gamma \curvearrowright X$ and $\Lambda \curvearrowright Y$ are orbit equivalent if E_Γ^X and E_Λ^Y are orbit equivalent.
- Similar notions apply for measure-preserving actions on probability spaces by using almost-everywhere bijections/isomorphisms.

Theorem (Feldman–Moore Theorem): Let E be a countable Borel relation. Then, there is a countable group Γ and a Borel action $\Gamma \curvearrowright X$ such that $E = E_\Gamma^X$. Moreover, the group Γ can be chosen such that xEy if and only if there exists $g \in G$ with $g^2 = e$ such that $g.x = y$.

Hyperfinite Equivalence Relations

Definition: A Borel relation E on a set X is called *smooth* if there is a Borel function $\phi: X \rightarrow Y$ with Y standard Borel satisfying xEy if and only if $\phi(x) = \phi(y)$.

Example: Every finite Borel equivalence relation is smooth. This is determined by setting a Borel linear order on X , then defining $\phi(x)$ to be the smallest element of $[x]$ for each $x \in X$.

Remark: If E is countable and smooth, then $\phi: X \rightarrow Y$ is countable-to-one, so its range is Borel. In particular, we may assume that ϕ is surjective.

Given any Borel equivalence relation E on X , we may define the space of equivalence classes X/E to have the quotient Borel structure by saying $A \subseteq X/E$ is Borel if and only if $\bigcup A$ is Borel in X .

Proposition: A countable Borel relation is smooth if and only if X/E is standard Borel.

Proof. If $\phi: X \rightarrow Y$ is a surjection witnessing the smoothness of E , we let $\bar{\phi}: X/E \rightarrow Y$ be the corresponding bijection. It suffices to show that $\bar{\phi}$ induces an isomorphism from the quotient Borel structure of X/E to the standard Borel structure of Y . Since ϕ is countable to one, ϕ maps Borel sets to Borel sets, so any E -invariant set B is Borel if and only if $\bar{\phi}(B) = \phi(B)$ is Borel.

In the reverse direction, if the quotient Borel structure on X/E is standard, then there is a countable family $\{B_n : n \in \mathbb{N}\}$ of subsets of the quotient Borel structure on X/E which separates points. However, such a family induces a Borel separating family on E ; by considering the functions $\{\chi_{B_n}\}_{n \in \mathbb{N}}: E \rightarrow 2^{\mathbb{N}}$, we obtain our desired result. $\square$

Definition: A complete section for E is a set $B \subseteq X$ such that B intersects every class of E .

A transversal for E is a complete section that intersects at exactly one point.

A selector for E is a function $f: X \rightarrow X$ with $\text{graph}(f) \subseteq E$ and $\text{im}(f)$ is a transversal.

Notice that if E is a Borel equivalence relation that admits a transversal $T \subseteq X$, then we may define a map $f: X \rightarrow X$ by taking $f(x) = [x] \cap T$. This is our desired map to witness smoothness of E , which is well-defined since T intersects $[x]$ at exactly one point, and has it such that xEy if and only if $y \in [x]$ if and only if $\phi(y) = [x] \cap T = \phi(x)$.

Proposition: If E is a smooth countable Borel relation, then E admits a selector.

Moreover, if all the classes of E are some fixed size $n \in \mathbb{N} \cup \{\infty\}$, then there is a sequence $(f_i)_{i < n}$ of Borel selectors for E whose images partition X .

Proof. Let $\phi: X \rightarrow Y$ witness the smoothness of E , and let $R \subseteq Y \times X$ be defined by yRx if and only if $\phi(x) = y$.

Since R is Borel and has countable sections, the Lusin–Novikov Uniformization Theorem gives a Borel map $f: \text{proj}_Y(R) \rightarrow X$ such that $\text{graph}(f) \subseteq R$. Here,

$$\text{proj}_Y(R) = \{y \in Y : \text{there exists } x \in X \text{ such that } (y, x) \in R\}.$$

Such a map is known as a Borel uniformization. The map $f \circ \phi: X \rightarrow X$ is our desired Borel selector.

Suppose all the classes of E are of size $n \in \mathbb{N} \cup \{\infty\}$. Then, there is a sequence $(f_i)_{i < n}$ of Borel uniformizations of R with disjoint ranges whose union is X . The maps $(f_i \circ \phi)_{i < n}$ is our desired sequence. $\square$

We say a countable Borel relation E on X is *hyperfinite* if there is an increasing sequence $\{F_n\}_{n \in \mathbb{N}}$ of finite Borel subequivalence relations of E .

Dye's Theorem

The goal is to show the following theorem.

Theorem (Dye's Theorem): All non-atomic probability measure-preserving ergodic actions of $\mathbb{Z}$ are orbit equivalent.

Equivalently, let (X, E, μ) and (Y, F, ν) be a.e. aperiodic hyperfinite Borel equivalence relations with invariant ergodic probability measures. Then, they are isomorphic.

Definition: Let E be a countable Borel relation. A FSR (finite partial subequivalence relation) is a finite Borel relation F defined on $\text{dom}(F) \subseteq X$ with $F \subseteq E$.

If $[X]^{<\infty}$ is the standard Borel space of finite subsets of X , and $[E]^{<\infty}$ is the Borel subset of $[X]^{<\infty}$ of pairwise E -related finite subsets, then if $\Phi \subseteq [E]^{<\infty}$, we say a FSR $F \subseteq E$ is Φ -maximal if

- for all $x \in \text{dom}(F)$, $[x]_F \in \Phi$;
- for all $S \in [X \setminus \text{dom}(F)]^{<\infty}$, we have that $S \notin \Phi$.

Lemma: Let E be a countable Borel relation, and let $\Phi \subseteq [E]$ be Borel. Then, E admits a Φ -maximal FSR.

Proof. Define a graph G on $[E]^{<\infty}$ by taking $(S, T) \in G$ if and only if $S \neq T$ with $S \cap T \neq \emptyset$.

Then, G admits a Borel $\aleph_0$ -coloring. That is, a map $c: [E]^{<\infty} \rightarrow I$ (where I is countable and discrete) with $(S, T) \in G$ implying that $c(S) \neq c(T)$. We will show this now.

Let $\{g_n : n \in \mathbb{N}\}$ be a sequence of Borel involutions whose graphs cover E (which exist by Feldman–Moore). Let $<$ be a Borel linear order of X . Let $S \in [E]^{<\infty}$, and let $\{x_i\}_{i < n}$ be an enumeration with respect to the corresponding ordering of S . Let $c(S)$ be the lexicographically smallest sequence $\{k_{ij} : i, j < |S|\} \subseteq \mathbb{N}$ such that $g_{k_{ij}}.x_i = x_j$. That is, we will let I be given by $\mathbb{N}^\infty$, which is the set of finitely supported natural number sequences.

Suppose toward contradiction that c is not a coloring. Then, there is $(S, T) \in G$ such that $c(S) = c(T)$. Since $c(S)$ and $c(T)$ are equal, it follows that the enumerations $\{x_i\}_{i < n}$ and $\{y_i\}_{i < n}$ are of equal cardinality. Since we assume that $S \cap T \neq \emptyset$, we may select $x_i = y_j$ without loss of generality to have $i < j$. Then, we have that

$i < j$ if and only if $x_i < x_j$ if and only if $x_i < g_{k_{ij}}(x_i)$ if and only if $y_j \leq g_{k_{ij}}(y_j)$ if and only if $y_j < y_i$ if and only if $j < i$. This gives that $i = j$. Yet, this means that for all $\ell < n$, we also have

$$\begin{aligned} x_\ell &= g_{k_{i\ell}}(x_i) \\ &= g_{k_{i\ell}}(y_i) \\ &= y_\ell, \end{aligned}$$

so that $S = T$, which contradicts the definition of $(S, T) \in G$.

Now, we may assume that c is a natural number coloring. Define a sequence $\{F_n : n \in \mathbb{N}\}$ of FSRs by saying $x F_n y$ if there exists $S \in \Phi$ such that $c(S) = n$, $x, y \in S$, and

$$S \cap \left(\bigcup_{m < n} \text{dom}(F_m) \right) = \emptyset.$$

Since $\text{dom}(F_m) \cap \text{dom}(F_n) = \emptyset$ for any $m \neq n$, it follows that

$$F = \bigcup_{n \in \mathbb{N}} F_n$$

is a FSR of E . Furthermore, $[x]_F \in \Phi$ for all $x \in \text{dom}(F)$.

Suppose $S \in [E]^{<\infty}$ with $S \in \Phi$. Then, $c(S) = n$ for some $n \in \mathbb{N}$, so that $S \cap \text{dom}(F_m) \neq \emptyset$ for some $m \leq n$, meaning that F is Φ -maximal. $\square$

Corollary: Suppose E is an aperiodic countable Borel equivalence relation. Then, E contains a finite Borel subequivalence relation all of whose classes are of cardinality n .

Proof. Let $E \subseteq X \times X$. By the above proposition, we may find a maximal FSR F whose classes are of cardinality n . Let

$$A = \{x \in X : [x]_E \text{ is the union of } F \text{ classes}\}.$$

Then, for each class C , we have $C \setminus \text{dom}(F)$ has at most $n - 1$ elements. Therefore, $X \setminus A$ is a smooth set (i.e., restricting E to $X \setminus A$ makes E smooth), and we may define a finite Borel subequivalence relation $E' \subseteq E|_{X \setminus A}$ with classes of cardinality n . Therefore, $F|_A \cup E'|_{X \setminus A}$ is our desired finite Borel subequivalence relation. $\square$

Theorem (Rokhlin Lemma): Let T be an aperiodic Borel automorphism, μ a probability measure on X , $n \geq 1$, and $\varepsilon > 0$. Then, there is a Borel complete section $A \subseteq X$ such that

- (i) $T^i(A) \cap T^j(A) = \emptyset$ for any $0 \leq i < j < n$;
- (ii) $\mu\left(X \setminus \bigcup_{i < n} T^i(A)\right) < \varepsilon$;
- (iii) $\mu(A) \leq 1/n$.

Proof. Suppose E_T is smooth. Then, there is a transversal S . If we let

$$B = \bigcup_{k \in \mathbb{Z}} T^{kn}(S),$$

then we observe that B is the countable union of images of S under Borel automorphism, and since S is a transversal and each $\ell \in \mathbb{Z}$ has a unique residue mod n . Thus, $B, T(B), \dots, T^{n-1}(B)$ partition X , giving (i), and have nonempty remainder, giving (ii). Finally, we may choose $A = T^i(B)$ to be such that $\mu(A) \leq 1/n$ to give the desired result.

Let $\{A_m : m \in \mathbb{N}\}$ be a vanishing sequence of markers for E_T . We will assume that E_T is not smooth. Therefore, we will assume that each A_m is doubly recurrent, in that for all $x \in A_m$, there exists $i < 0 < j$ such that $T^i(x), T^j(x) \in A_m$ (i.e., none of the sequence of markers is a transversal).

Given a doubly recurrent Borel complete section $B \subseteq X$, one may define

$$d_l(x, B) = \min\{k \in \mathbb{N} : T^{-k}(x) \in B\}$$

$$d_r(x, B) = \min\{k \in \mathbb{N} : T^k(x) \in B\}.$$

Let B_m be defined by the set of all x such that either $d_l(x, A_m) < n$ or $d_r(x, A_m) < n$. Since the A_m are vanishing, it follows that the B_m are vanishing as well. Therefore, we may find $m \in \mathbb{N}$ such that $\mu(B_m) < \varepsilon$.

For each $k < n$, say $x \in C_k$ if and only if $d_l(x, A_m) \equiv k$ modulo n and $d_r(x, A_m) \geq n$. Since these sets are pairwise disjoint, it follows that $\mu(C_k) \leq 1/n$ for some $k < n$, so $A = C_k$ satisfies (i), with (ii) following from the fact that $X \setminus B_m \subseteq \bigcup_{i < n} T^i(A)$. $\square$

References

- [Kec95] Alexander S. Kechris. *Classical Descriptive Set Theory*. Vol. 156. Graduate Texts in Mathematics. Springer-Verlag, New York, 1995, pp. xviii+402. ISBN: 0-387-94374-9. DOI: [10.1007/978-1-4612-4190-4](https://doi.org/10.1007/978-1-4612-4190-4). URL: <https://doi.org/10.1007/978-1-4612-4190-4>.
- [KM04] Alexander S. Kechris and Benjamin D. Miller. *Topics in Orbit Equivalence*. Vol. 1852. Lecture Notes in Mathematics. Springer-Verlag, Berlin, 2004, pp. x+134. ISBN: 3-540-22603-6. DOI: [10.1007/b99421](https://doi.org/10.1007/b99421). URL: <https://doi.org/10.1007/b99421>.
- [KL16] David Kerr and Hanfeng Li. *Ergodic Theory*. Springer Monographs in Mathematics. Independence and dichotomies. Springer, Cham, 2016, pp. xxxiv+431. ISBN: 978-3-319-49845-4; 978-3-319-49847-8. DOI: [10.1007/978-3-319-49847-8](https://doi.org/10.1007/978-3-319-49847-8). URL: <https://doi.org/10.1007/978-3-319-49847-8>.